Training–Rollout Mismatch in Low-Precision LLM Reinforcement Learning: A Survey

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Abstract: Low-precision rollout generation changes more than the cost of reinforcement learning: it can change the distribution that supplies the learner's training data. The resulting training–rollout mismatch is difficult to characterize because quantization interacts with execution engines, policy updates, decoding decisions, and the evolving distribution of prefixes. This survey organizes the problem around three questions: what differs between learner and sampler, how those differences affect learning, and where interventions act. We distinguish execution mismatch from policy staleness and separate numerical agreement, optimization stability, and task quality. A component-level taxonomy groups interventions into numerical design and execution alignment, and learning under residual mismatch. These groups cover computation alignment, quantization-error control and parameter-update preservation, gradient-estimation correction, policy-update control, and optimization-schedule adaptation. We synthesize a curated set of twenty focal studies, distinguish direct low-precision evidence from general mismatch studies and boundary cases, and compare assumptions, information requirements, and implementation costs. The literature supports a conditional view: neither a shared precision format nor a small average probability gap is sufficient to establish effective learning. We conclude with open problems in diagnosis, method composition, asynchronous generation, and evidence quality. This is a literature synthesis; it introduces no new benchmark results.

Keywords: quantized rollout; low-precision reinforcement learning; training–inference mismatch; importance sampling; quantization-aware training.

Literature snapshot: October 19, 2026.

Survey resources: https://github.com/ZunhaiSu/awesome-low-precision-rl

1. Introduction and Scope

1.1. Why Low-Precision Rollouts Complicate Policy Optimization

The sampler in language-model reinforcement learning serves two roles: it executes a policy, and it produces the observations from which the next policy is learned. An approximation that is acceptable for a fixed inference workload can therefore have a different consequence inside a training loop. Its outputs influence rewards, advantages, subsequent parameter updates, and future sampling. Low precision makes this feedback especially visible, although numerical differences between training and serving engines can occur even when the nominal precision is the same.

This creates a distinctive optimization problem. Reducing rollout cost is useful only if the resulting training process still produces a useful policy within the available budget. A method can improve probability agreement yet suppress informative updates; another can preserve task performance while leaving measurable numerical disagreement. These outcomes should not be collapsed into a single notion of "alignment." The practical question is which discrepancies matter for learning, which interventions address them, and what those interventions cost.

The recent literature approaches this question through several interfaces. FlashRL motivates correction using actual rollout probabilities; Jet-RL changes the numerical paths shared by learner and sampler; later studies intervene in quantization, clipping, gradient estimation, or local update dynamics [1, 3]. Their contributions are easiest to compare when the unit of analysis is the *method component*, rather than the paper title or precision format. An integrated system may contain both an execution intervention and an optimization intervention, neither of which should disappear behind a single label.

Figure 1 previews the two intervention interfaces. The distinction is operational: numerical methods change the computation that realizes a policy, while learning methods change the update computed from sampled data.

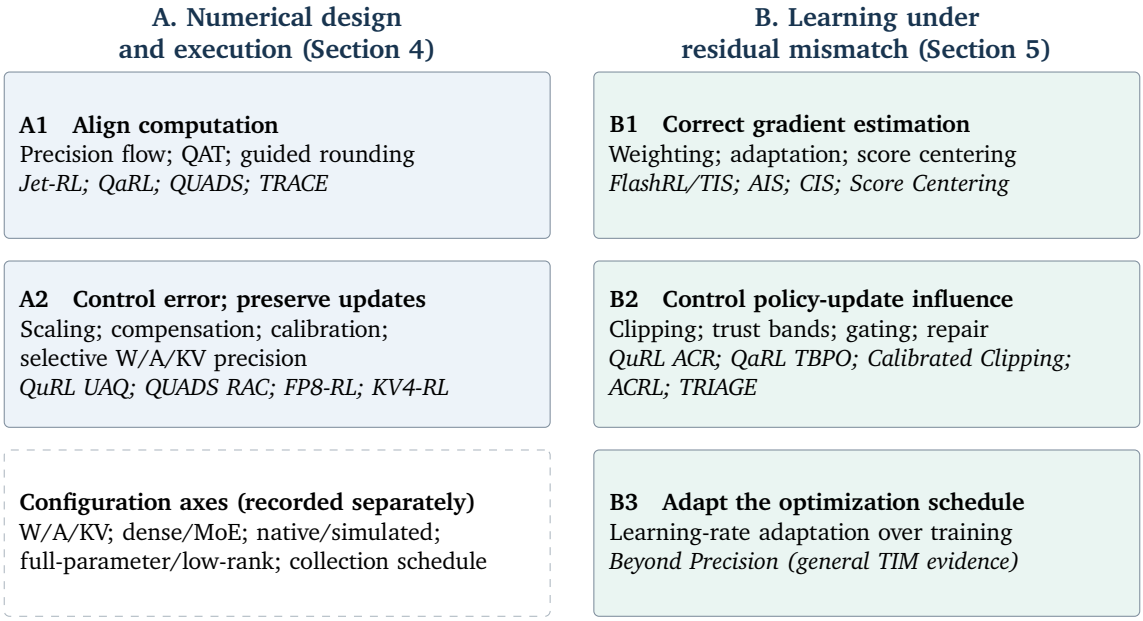


Figure 1: Road map of the component taxonomy. Integrated studies can contribute components to both columns; configuration labels describe the experimental regime separately. The detailed taxonomy and evidence boundaries are developed in Sections 2, 4, and 5.

1.2. Scope, Related Surveys, and Review Protocol

Our focus is numerical and execution mismatch in LLM reinforcement learning, with emphasis on quantized rollout and closely related low-precision training configurations. We include asymmetric settings, such as a higher-precision learner with a lower-precision sampler, and settings in which both sides use low precision. Weight, activation, and key–value (KV) cache quantization are distinct numerical targets. Policy staleness is discussed where it interacts with execution mismatch, not treated as a synonym for it.

The focal corpus contains twenty research contributions: thirteen method or system studies, six studies used primarily for formulation or general mismatch analysis, and one exploration-oriented boundary case. These are *editorial roles*, not quality tiers or mutually exclusive evidence types. For example, MiniRL supports our formulation while also containing a direct FP8-rollout setting [14]. We count a conference version and its associated preprint as one contribution. Appendix A supplies the mapping; the companion data file records source and version information.

This is a focused narrative survey built from a curated seed collection and targeted source checking. It is not presented as an exhaustive systematic review: there is no claimed database-wide screening count or reproducible recall estimate. The manuscript reflects sources available in the working collection on October 19, 2026. General post-training quantization, pretraining-only low-precision methods, and speculative-draft acceleration are outside the method taxonomy unless they directly illuminate the learner–sampler interface. Search coverage before this snapshot is not reconstructed.

Broader surveys cover low-precision LLM training, rollout strategies, and reinforcement learning for reasoning models [24–26]. Our emphasis is narrower: the relationship between actual sampling, numerical execution, and the update computed from sampled data. The exposition follows the component taxonomy developed below; numerical formats and experimental regimes are retained as separate comparison axes.

1.3. Research Questions and Contributions

Four questions guide the review. **RQ1:** How should execution mismatch be separated from policy changes and decoding differences? **RQ2:** Through which mechanisms can disagreement distort estimation or optimization? **RQ3:** What do numerical and learning interventions directly modify? **RQ4:** What conclusions remain defensible after accounting for differences in experimental setting and system cost?

The survey makes three contributions. First, a policy notation separates numerical execution, decoding, and version lag, and exposes the difference between conditional and response-level correction. Second, a component taxonomy compares numerical coupling, fidelity control, estimator correction, update control, and schedule adaptation without assigning integrated systems to a single method box. Third, a source-linked evidence synthesis relates reported outcomes to their information requirements, evaluation rules, and costs. Sections 2 and 3 define the problem and its consequences. Sections 4 and 5 compare interventions. Section 6 summarizes published evidence without constructing a cross-paper leaderboard. Section 7 derives research questions from the remaining gaps.

2. Problem Formulation and Taxonomy

2.1. Learner, Behavior, Proximal, and Reference Policies

Let x be a prompt, $y_{1:T}$ a response, and $h_t = (x, y_{<t})$ its prefix. We distinguish a parameter state from the execution path used to evaluate it. The current learner policy is $p_\theta(a \mid h)$. At a sampling snapshot θ_k, the same learner-side path defines $p_k(a \mid h)$. The actual behavior policy is $q_k(a \mid h)$, including the rollout engine, numerical representation, and declared sampling transformation. Temperature or vocabulary truncation must therefore be recorded when interpreting behavior probabilities. A returned raw-model log-probability is not automatically the log-probability after the sampling transformation. The notation below conditions on a fixed, documented collection protocol. Under deterministic execution and version schedules, cache state is determined by the generated history even when it contains computations from earlier versions; the subscript k is shorthand for this specified rollout path.

A proximal policy p_{prox} supplies the denominator of a policy-update constraint. A reference policy p_{ref} supplies a regularization target. These are different roles: a frozen pre-update learner may be proximal, while a much earlier policy is the KL reference. Neither is automatically the policy

that generated every token. PPO and GRPO provide familiar contexts for these distinctions [21, 22]; mismatch-aware formulations require them to remain explicit [1, 14].

Table 1: Policies and quantities used throughout the survey. Execution paths are part of policy identity.

Symbol	Meaning
p_θ	Current learner policy under its specified execution path.
p_k	Learner-side policy evaluated at sampling snapshot θ_k.
q_k	Actual behavior policy that sampled the token.
p_{prox}	Policy used to define the proximal update ratio.
p_{ref}	Policy used for KL regularization; not the behavior denominator.
Δ_k	Same-snapshot log-probability difference, $\log p_k - \log q_k$.
d_θ	Current learner–behavior gap, $\log p_\theta - \log q_k$.
A_t	Advantage or other policy-gradient learning signal.
$s_\theta(a,h)$	Score vector, $\nabla_\theta \log p_\theta(a \mid h)$.

2.2. Execution Mismatch versus Policy Staleness

At a fixed prefix and token with positive relevant probabilities,

$$\frac{p_\theta(a \mid h)}{q_k(a \mid h)} = \underbrace{\frac{p_\theta(a \mid h)}{p_k(a \mid h)}}_{\text{parameter change}} \underbrace{\frac{p_k(a \mid h)}{q_k(a \mid h)}}_{\text{same-snapshot execution/decoding difference}}. \tag{1}$$

This is an algebraic identity, not an independence assumption. The first factor includes updates since collection and, in asynchronous settings, version lag. The snapshot policy p_k is a conceptual same-parameter reference; an implementation need not retain it explicitly to use a different proximal policy. The second isolates numerical execution only when decoding rules are controlled. If temperature or top-p filtering differs, it contains a decoding component as well. In partial rollouts, the sampling index may be token-dependent, $k = k(t)$. Moreover, carrying a cache across synchronization can preserve computations from earlier parameter states. In such an implementation, the current checkpoint index alone does not specify the behavior policy; cache provenance or the realized sampling probabilities are also needed. This is a modeling consideration for partial collection, rather than a claim that every partial-rollout implementation retains stale caches.

Probability ratios also require support conditions. A token observed from q_k has a defined denominator, but that alone does not establish that q_k covers every action with positive target probability. Truncation sampling can violate this condition. Similarly, a token-level change of measure is not automatically a correction for the prefix distribution. At fixed h,

$$\mathbb{E}_{a \sim q_k(\cdot \mid h)}\left[\frac{p_\theta(a \mid h)}{q_k(a \mid h)} f(a,h)\right] = \mathbb{E}_{a \sim p_\theta(\cdot \mid h)}[f(a,h)] \tag{2}$$

holds under support and integrability assumptions. It says nothing by itself about whether the outer distribution over h is the desired one. It also does not undo changes to group-relative advantages induced by collecting different responses.

For comparison, the full-response likelihood ratio is

$$W_\theta(y \mid x) = \frac{P_\theta(y \mid x)}{Q(y \mid x)} = \prod_{t=1}^{T} \frac{p_\theta(y_t \mid h_t)}{q_{k(t)}(y_t \mid h_t)}. \tag{3}$$

Here P_θ and Q are the induced response laws, and the denominator factors must be their actual conditional probabilities under the documented protocol. Unobserved execution randomness requires marginalizing that state or extending the likelihood to the corresponding execution events. The two response laws use a common stopping convention; an end-of-sequence decision is included when applicable. For a fixed behavior law and a parameter-independent reward R, the usual likelihood-ratio identity gives

$$\nabla_\theta \mathbb{E}_{P_\theta}[R] = \mathbb{E}_Q\left[W_\theta R \sum_{t=1}^{T} s_\theta(y_t, h_t)\right] \tag{4}$$

under support, integrability, and differentiation assumptions. The product illustrates why exact response-level correction can have high variance; multiplying only the current token ratio into an arbitrary sampled advantage is a different estimator. Nor does Equation (4) directly justify reusing group-normalized advantages computed under Q. Score Centering explicitly distinguishes sequence-level importance sampling from practical token-level corrections [10].

2.3. A Component-Level Taxonomy of Interventions

We group components by the object they directly modify. **A: numerical interventions** alter representation or execution. A1 matches computation between learner and sampler; A2 controls quantization error or preserves parameter updates. **B: learning interventions** alter how sampled data produce updates. B1 corrects gradient estimation; B2 controls the acceptance, magnitude, or direction of policy updates; B3 adapts the optimization schedule.

These are organizing categories, not disjoint mathematical sets. Their boundaries answer an operational question: if the component is removed while the rest of the system is held fixed, which computation or update rule changes? An estimator correction changes an update, and an execution alignment can improve quantization fidelity. The primary assignment follows the component's operative rule and design objective. In particular, a multiplicative weight is not sufficient to identify importance sampling. We distinguish a weight constructed to correct a distributional estimation problem from a weight constructed to control how an update changes disagreement. B3 occupies a short subsection because its direct low-bit evidence in this corpus is limited.

Table 2: Five intervention classes. These categories describe operative components, not exclusive paper types.

Class	Direct object	Representative components	Important residual question
A1	Shared computation	Precision flow; aligned QAT; guided rounding	Which interfaces remain unmatched?
A2	Quantization fidelity	UAQ; residual compensation; calibration; selective precision	Are useful parameter changes visible?
B1	Gradient estimator	TIS; AIS; CIS; Score Centering	Which expectation is corrected, and with what bias?
B2	Update influence	ACR; TBPO; calibrated clipping; direction-aware controls	Which updates are retained or suppressed?
B3	Optimization schedule	Mismatch-informed LR adaptation	Is stability obtained at comparable learning progress?

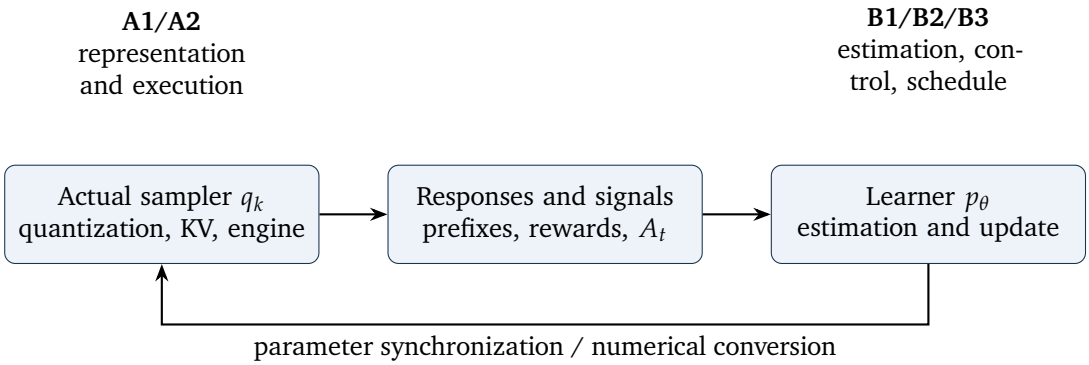


Figure 2: The feedback loop reviewed in this survey. Numerical interventions may affect either execution endpoint; learning interventions act on the update. Arrows express data flow, not a claim that interventions must be applied in a particular order.

2.4. Configurations and Evidence Labels

The taxonomy should not absorb configuration axes. W/A/KV precision, dense versus mixture-of-experts (MoE) architecture, low-rank versus full-parameter updates, synchronous versus partial collection, and native kernels versus quantize–dequantize simulation are recorded separately. "FP4 training" does not specify which operations use FP4 or whether master weights, backward computation, and optimizer states remain at higher precision.

We likewise separate a paper's role in this survey from the evidence it contains. A formulation paper can include direct quantized-rollout experiments; a quantized system paper can reuse a correction developed elsewhere. This distinction prevents two errors: dismissing relevant experiments because a paper is background reading, and attributing an adopted correction to the system that integrates it.

3. Mechanisms: From Numerical Discrepancy to Learning Dynamics

3.1. Sources and Propagation of Execution Discrepancy

A numerical policy is determined by more than its checkpoint. Scale selection, rounding, accumulation, kernel choice, batch shape, and the treatment of cached states can alter the output distribution. Sharing a nominal format therefore does not imply sharing a computation. FP16-based mismatch analysis provides a useful precision-selection comparison, while R3 exposes the importance of matching discrete routing decisions in MoE execution [15, 16]. Neither should be treated as evidence that every routing discrepancy originates in quantization.

Two measurement settings must remain separate. *Shared-prefix comparison* holds the token history fixed and compares conditional distributions. *Free-running comparison* allows sampling differences to change the histories visited. The first is valuable for isolating a numerical interface; the second measures behavior after divergence has propagated through generation. A favorable shared-prefix statistic is not, on its own, a guarantee of similar trajectory-level learning signals.

Discretization introduces a further asymmetry. An update can be visible in the learner's parameter space but fail to change the sampler's quantized representation. Conversely, a small change near a rounding boundary can abruptly change a codeword. QuRL studies the former through update-

aware quantization; TRACE targets inconsistency in rounding outcomes [5, 13]. The relevant object is therefore not only error magnitude, but its relationship to the evolving parameters and the discretization boundaries.

3.2. Distorted Estimation and Update Decisions

A probability difference and an importance ratio encode different information. Writing a learner probability as $p = q + \delta$ gives $p/q = 1 + \delta/q$: the same absolute perturbation has a larger relative effect when q is small. This elementary observation motivates reporting tail behavior and confidence-conditioned statistics rather than only mean absolute probability error. CIS makes confidence dependence explicit in its correction design [20].

Estimation can also acquire a systematic drift. For a normalized differentiable learner distribution, $\mathbb{E}_{a\sim p_\theta}[s_\theta(a,h)] = 0$. Under behavior sampling, the corresponding expectation need not vanish. At fixed h,

$$\mathbb{E}_q[A s_\theta] = \mathrm{Cov}_q(A, s_\theta) + \mathbb{E}_q[A]\mathbb{E}_q[s_\theta]. \tag{5}$$

The identity describes a conditional estimator; it does not assume that centering advantages across an entire prompt group centers them at every prefix. Score Centering uses this distinction to motivate an additive correction [10].

Mismatch additionally affects decisions made by a surrogate objective. In a clipped policy-gradient objective, changing a ratio can change which branch is active and therefore whether a token contributes gradient. Thus, a perturbation can matter through a threshold crossing even when its absolute magnitude appears modest. The full-pipeline FP8 study isolates a failure involving excessive clipping of negative-advantage tokens [11]. This is a different intervention target from merely decreasing the variance of a sampled importance weight.

The sign dependence can be seen without assuming a particular quantizer. Let $\ell(r, A) = \min\{rA, \mathrm{clip}(r, 1-\epsilon, 1+\epsilon)A\}$, with $0 < \epsilon < 1$ and fixed A. Away from the two boundaries,

$$\frac{\partial \ell}{\partial r} = \begin{cases} A\,\mathbf{1}\{r < 1+\epsilon\}, & A > 0, \\ A\,\mathbf{1}\{r > 1-\epsilon\}, & A < 0, \\ 0, & A = 0. \end{cases} \tag{6}$$

Thus the active clipping side depends on the learning signal. For an illustrative objective that clips the combined learner–behavior ratio, a same-snapshot ratio of 0.7 with $\epsilon = 0.2$ already suppresses a negative-advantage token, even before a learner update. This is not the decoupled surrogate in Section 5.1; the example explains why naming the clipped ratio is essential to interpreting a failure or a repair.

Figure 3 makes this sign dependence explicit for the standard surrogate. It also shows why clipping the combined learner–behavior ratio can suppress an update that a proximal ratio would still permit.

3.3. Feedback across Optimization Steps

The learning loop turns a local discrepancy into a dynamic process. A sampled response changes the learner; the learner is synchronized back to the sampler; the next rollout is then generated from an updated numerical realization. Stability depends on this feedback, not solely on whether one frozen batch has a small error.

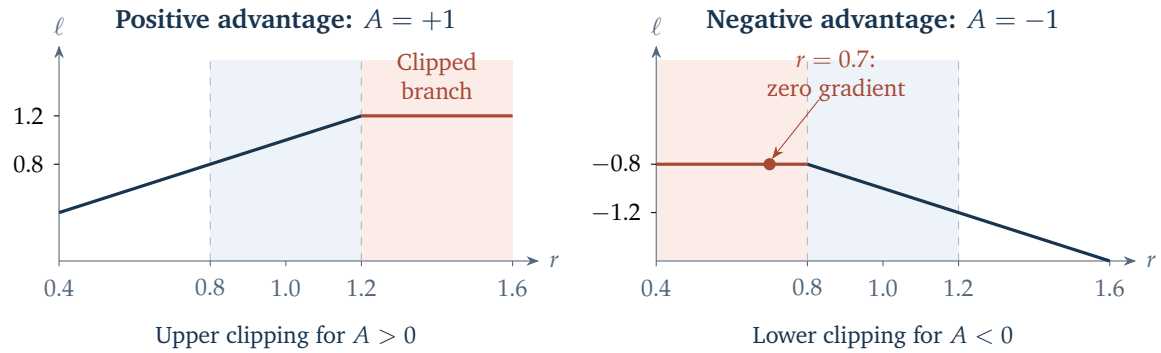


Figure 3: Analytical illustration of Equation (6) with $\epsilon = 0.2$ and $A = \pm 1$. Blue shading marks $[0.8, 1.2]$; red shading marks the flat, clipped branch. A ratio of 0.7 lies on the zero-gradient branch for negative advantage. The relevant interpretation depends on which policy ratio the objective clips.

Direction matters because two updates with the same gap magnitude can move learner and behavior probabilities differently. In a simplified local picture, reducing a learner probability that is already below the behavior probability can enlarge their log gap. ACRL and TRIAGE use update-direction information in their controls [9, 12]. For a shared-parameter network, however, one token's update changes other probabilities too. The simple sign argument is explanatory intuition, not a universal stability theorem; formal conditions and empirical scope must be taken from the individual method.

Optimization scale adds another feedback channel. Beyond Precision studies the relationship between mismatch and optimization dynamics and proposes response-length-triggered learning-rate decay [18]. Its role here is to show why numerical observations should be interpreted together with update schedules. General mismatch experiments do not, by themselves, establish the effectiveness of that schedule under aggressive FP4 execution.

3.4. Diagnosing the Effects of Mismatch

A diagnosis must specify both the intervention and the counterfactual. Replacing an inference engine can simultaneously change arithmetic, batching, memory use, and generated trajectories. A resulting quality difference cannot automatically be assigned to the precision format. VeXact constructs a zero-mismatch reference to help separate execution mismatch from other training effects [19]. Such a reference is useful because it changes what can be attributed, not because it is necessarily the preferred accelerated deployment configuration.

A coherent evidence chain asks whether an execution change altered the measured discrepancy, whether the discrepancy affected the update, and whether the update affected learned quality. Ablations may support some links without identifying the entire chain. This is why we retain paper-specific conditions when describing instability and avoid turning a result from one architecture and horizon into a universal failure law.

Box 1. Does smaller mismatch always mean better learning?

QeRL combines low-precision execution, low-rank adaptation, and adaptive quantization noise, making exploration an explicit part of its setting [2]. It is a boundary case for this survey, not a method for eliminating mismatch. Its results motivate separating numerical agreement from task utility, but do not imply that arbitrary cross-engine error is beneficial. A useful comparison must ask what changed in the sampling distribution, what adaptation capacity the learner had, and whether the gain persists under matched evaluation.

Synthesis. Mismatch is not one scalar defect. Conditional probability error, trajectory selection, estimator drift, threshold decisions, and cross-step feedback are distinct objects. A method should be assessed against the object it actually changes.

4. Numerical Design and Execution Alignment

Numerical interventions change the policy-producing computation. Their immediate target is a representation, numerical transformation, or execution decision, rather than an importance weight or policy-gradient objective. We distinguish alignment to the other endpoint from control of quantization fidelity. The distinction is operational: reducing a sampler's approximation error can help without requiring it to reproduce the learner's path, whereas matching two paths need not make either path more accurate relative to a higher-precision model.

4.1. Aligning Learner and Rollout Computation

From matching formats to matching paths. Jet-RL organizes learner and rollout execution around a unified FP8 precision flow, including consistent granularity at corresponding interfaces [3]. Its relevance is the shift from describing endpoints by a bit width to describing how precision propagates through operators. This is an alignment intervention, although a common format should not be equated with universal bitwise identity across engines.

Training against the rollout representation. QaRL brings learner-side forward computation closer to the quantized rollout path through rollout-aligned quantization-aware training (QAT) [6]. The comparison of interest is the representation on which the update is computed. QAT and native low-precision kernels should still be documented separately: matching their intended quantizer does not ensure identical execution of every operator. QaRL's additional TBPO objective is discussed in Section 5.2, rather than folded into its numerical contribution.

Aligning only the parts that can be shared. QUADS uses an asymmetric learner configuration: weights are fake-quantized while learner activations remain at higher precision [8]. This makes weight-path matching distinct from activation-error treatment. The design illustrates why symmetrical bit widths are not the only possible alignment strategy. A useful classification must permit a method to match one interface and compensate another, instead of requiring the whole system to be uniformly low precision.

Sharing decisions rather than specifications. TRACE records compact rollout-side quantization information and uses it to guide learner-side FP4 rounding [13]. Sharing a quantizer specification and sharing its realized decisions are different operations. The latter can resolve inconsistencies caused by inputs landing on opposite sides of a rounding boundary. TRACE's local rounding argument should nevertheless be distinguished from a guarantee of monotonic reduction in full-network policy divergence.

This progression should not be read as a universal superiority ordering: each design couples the endpoints at a different interface. These interventions can be compared by the information they require. A common precision flow shares a design; aligned QAT shares or emulates a representation; guided rounding transfers realized numerical information. Routing replay, discussed as background through R3, follows a related principle for discrete execution choices [15]. Greater coupling can improve agreement at a specific interface, while also increasing dependence on matching token positions, module identities, and parameter versions. Those dependencies are part of the method, not merely implementation details.

4.2. Controlling Quantization Error and Preserving Parameter Updates

Preserving changes that training intends to make. For an illustrative deterministic quantizer Q, a parameter update may satisfy

$$Q(w + \delta w) = Q(w), \tag{7}$$

even when $\delta w \neq 0$. Whether this occurs depends on the distance to a quantization boundary and on scale behavior, not only on the size of the update. QuRL's Update-Aware Quantization uses invariant scaling to alter the relation between quantization resolution and parameter-update magnitude [5]. It belongs here because its direct operation changes numerical parameterization. Its ACR component instead changes clipping and belongs in Section 5.2.

Compensating residual activation error. QUADS complements learner-side weight alignment with Residual Activation Compensation on the rollout side [8]. This is an error-control component within a dual-side system. Separating it from the QAT component makes the central comparison explicit: one operation matches a path, while the other reduces the error that remains in another path. Performance of the combined system does not identify either component's contribution without its corresponding ablation.

Calibrating changing distributions. FP8-RL integrates blockwise W8A8 rollout, scale recalibration for FP8 attention/KV execution, and dynamic weight synchronization [4]. Its quantization and calibration components belong in A2; its adopted TIS/MIS correction is cross-referenced in B1. The system contribution is their integration under changing RL parameters. The v2 study also explores end-to-end FP8, so learner precision must be recorded per experiment rather than assigned once to the entire paper.

Allocating precision selectively. KV4-RL uses INT4 KV storage with bounded INT8 protection of attention-sink and recent positions [17]. This changes where numerical fidelity is preserved, instead of uniformly increasing cache precision. The retained TIS component does not remove the need to study sampler fidelity. Its workload setting is also specific: colocated, quota-based partial collection

lets responses span collection boundaries while generation and optimization alternate. It should not be described as simultaneous learner–sampler execution.

The common question across A2 is where precision expenditure changes the effective learning process. Scale calibration, parameter rescaling, residual compensation, and positional protection are different mechanisms, even though all can improve a low-precision system. The distinction also explains why KV is an auxiliary label rather than a separate top-level method family: KV states can be calibrated, selectively protected, or aligned through shared information.

4.3. Residual Mismatch and Limits of Alignment

Alignment has a scope. It may hold for weights but not activations, for a quantizer but not surrounding kernels, or for selected states rather than every module. A paper's guarantee or measurement should therefore be attached to the relevant interface. The phrase "matched execution" is incomplete without specifying what is matched and under which parameter snapshot and input history.

Three implications follow. First, alignment is compatible with residual learning correction: addressing one source does not remove unrelated discrepancies. Second, two approximations can agree while both differ from a chosen higher-precision reference, so agreement and numerical accuracy are separate axes. Third, correcting a local computation does not necessarily recover trajectories already excluded by earlier sampling decisions. These implications motivate learning-side interventions without implying that numerical alignment must always precede them.

Synthesis. A1 asks how the two endpoints can share computation; A2 asks where numerical fidelity and update visibility should be preserved. Comparing them requires naming the protected interface, the remaining mismatch, and the information exchanged.

5. Learning under Residual Mismatch

Learning interventions accept that the rollout data and learner computation may differ. Their goals range from adjusting an estimator to controlling harmful updates or changing the optimization schedule. They should not all be evaluated as approximations to the same unbiased gradient: some deliberately modify the update to trade bias, variance, or stability against one another.

5.1. Correcting Gradient Estimation

Multiplicative correction and truncation. Importance sampling supplies a conditional change of measure under the assumptions of Equation (2). Practical surrogate objectives usually distinguish the mismatch correction from the proximal update ratio. With frozen p_{prox} and recorded behavior q, define

$$c_t = \frac{p_{\text{prox}}(a_t \mid h_t)}{q(a_t \mid h_t)}, \qquad r_t(\theta) = \frac{p_\theta(a_t \mid h_t)}{p_{\text{prox}}(a_t \mid h_t)}. \tag{8}$$

A representative decoupled surrogate is

$$\widehat{L}(\theta) = \frac{1}{N} \sum_t \text{sg}(\bar{c}_t) \min[r_t(\theta) A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) A_t], \tag{9}$$

where $\bar{c}_t$ is a possibly truncated correction, the sampled advantages are held fixed, and sg denotes no differentiation through the correction weight. This is a schematic comparison device, not a claim that every reviewed method implements Equation (9) identically. It makes two distinct interventions visible: clipping c_t and clipping the update ratio r_t.

FlashRL's TIS provides a basic bounded correction using actual rollout probabilities [1]. MiniRL exemplifies the mismatch-aware distinction between behavior and proximal policies underlying this notation [14]. FP8-RL adopts token-level TIS/MIS variants, while KV4-RL retains TIS alongside cache protection [4, 17]. These uses contribute system evidence without creating additional estimator families. Truncation limits extreme influence but changes the expectation. A stable truncated objective should therefore not be described as exact importance sampling merely because its denominator comes from the sampler.

Adapting the amount and location of correction. AIS uses observable mismatch diagnostics to interpolate between uncorrected and fully importance-weighted gradients [7]. Its adaptive quantity is the estimator mixture, not a precision schedule. CIS instead derives a confidence-dependent importance-weight cap from a discrepancy representation [20]. These approaches differ in what they adapt: overall correction strength versus the location of truncation across token confidence. CIS is used here as general TIM evidence; its BF16 cross-engine experiments are not direct FP4 validation.

Additive score correction. Score Centering replaces the score at a prefix with

$$\widetilde{s}_\theta(a,h) = s_\theta(a,h) - \mathbb{E}_{b\sim q(\cdot|h)}[s_\theta(b,h)]. \tag{10}$$

With the exact expectation, its mean under q vanishes and the drift term in Equation (5) is removed [10]. The remaining covariance is still measured under q, so this is not full recovery of an on-policy gradient. The reported implementation uses sampler top-k log-probabilities and models the omitted tail using the learner distribution; its default is $k = 128$ [10]. The resulting approximation is more specific than simply discarding the tail. Its coefficients are detached when forming the correction, and its fidelity depends on the reconstructed behavior distribution.

These estimator arguments assume that the reported score follows the stated learner computation. Surrogate gradients used through quantizers must be documented when interpreting identities involving derivatives. Comparison therefore requires specifying the target quantity, correction operator, information requirement, and unresolved error. Ratio-based correction needs probabilities of sampled actions; exact score centering involves a behavior-distribution expectation. Clipping and approximating these quantities change what can be guaranteed. No single label such as "bias correction" replaces these distinctions.

5.2. Controlling the Influence of Policy Updates

Restoring or reshaping update constraints. QuRL's Adaptive Clipping Range adjusts the update boundary in response to the relationship between proximal and behavior probabilities [5]. QaRL's TBPO combines negative-sample treatment with sequence-level trust-band control [6]. The full-pipeline FP8 method first matches the lower clipping quantile on negative-advantage tokens to a BF16 reference, then adjusts the upper boundary to rebalance positive and negative contributions [11]. Periodic reference passes amortize the calibration cost. All three act on the update constraint,

but use different signals and aggregation levels. “Calibration” in the last method refers to objective boundaries, not quantizer scales.

These interventions should be compared through the decisions they alter. Does the rule retain a token that would otherwise be clipped, reject an entire response, or change the balance of positive and negative updates? A wider acceptance region is not intrinsically better: its value depends on which updates enter and how their influence is bounded. Likewise, fewer clipped tokens can reflect recovered learning signal or inadequate control; the count alone cannot distinguish the two.

Controlling disagreement through update direction. ACRL uses advantage-guided probability adjustment, with response-level discrepancy relative to a reference controlling token reweighting [9]. Its objective is regulation, not unconditional minimization of discrepancy. TRIAGE instead combines segment-local risk information, direction-selective gating, rebalancing, and bounded repair in native NVFP4 execution [12]. Their common placement in B2 reflects the role of the update rule, not equivalence of their risk models or experimental settings.

The granularity of control matters. A response-level statistic aggregates evidence across many tokens; a segment statistic can expose local concentration; a token rule is directly selective but may be sensitive to noise. Aggregation trades localization against statistical reliability. Finer granularity improves localization only when the resulting diagnostic remains reliable enough to guide the update.

5.3. Adapting the Optimization Schedule

A third intervention changes the scale of learning over time. Beyond Precision motivates learning-rate scheduling through the evolution of optimization and mismatch [18]. This is complementary to changing a numerical path or a token weight: it modifies the trajectory of parameter updates themselves.

The reviewed evidence supports schedule adaptation as a complementary direction, but does not yet establish a broad family of low-bit-specific methods. Slower updates can also make a run appear stable by reducing learning progress. Any practical interpretation should therefore distinguish avoiding divergence from reaching comparable quality within a comparable budget. Extending general TIM schedule findings to aggressive quantization remains a research question.

5.4. Compatibility, Bias, and Failure Modes

A component taxonomy makes composition visible, but does not prove that arbitrary combinations work. QaRL, QuRL, and QUADS contain published combinations within one design. Score Centering also describes composition with importance weighting by centering the weighted score [5, 6, 8, 10]. These are stronger grounds for discussing compatibility than simply placing two methods in different boxes.

Three checks organize the comparison. *Semantic compatibility* asks whether components use the same behavior policy, parameter snapshot, and ratio definitions. *Statistical compatibility* asks whether multiple truncations or masks suppress the same signal and whether a centering term matches the estimator being centered. *Computational compatibility* asks whether required probabilities, quantization information, or reference passes can be obtained at acceptable cost. The third check is evaluated through published system evidence in Section 6.

A failure analysis becomes informative when it names what an intervention leaves untouched. A conditional correction does not change collected prefixes. A clipping rule does not reproduce the sampler's arithmetic. Numerical alignment does not remove policy lag. A schedule does not establish the correctness of a behavior denominator. These observations guide method selection more reliably than a generic claim that one method "solves mismatch."

Synthesis. The relevant distinction is between changing the computation, changing the estimator, and changing the permitted update. Methods can share an implementation primitive while pursuing different targets.

6. Synthesis of Findings and Practical Trade-offs

This section identifies what can be concluded from published studies and what depends on their settings. A numerical result is informative only together with its model, numerical path, training budget, baseline, and evaluation policy. We therefore compare mechanisms and evidence conditions instead of ranking unrelated speedup claims.

6.1. Coverage of Existing Studies

The corpus spans several configurations that should remain distinct. Asymmetric learner–sampler precision emphasizes a gap between numerical representations. Low precision on both sides emphasizes the consistency of their execution paths and the behavior of the resulting update. KV-focused work changes a state-storage interface, while weight or activation quantization changes other parts of computation. Low-rank adaptation and full-parameter RL additionally differ in which degrees of freedom can respond to quantization.

The autoregressive equations in Sections 2–5 describe token-generation paths. AIS also evaluates diffusion-based LLaDA alongside autoregressive Qwen models [7]. Its denoising experiments extend the empirical scope of the source, but require a generation-process-specific likelihood interpretation; they are not additional validation of the autoregressive factorization as written here. We use its autoregressive evidence for direct comparison in that formulation.

Architecture and collection protocol further condition conclusions. MoE execution introduces discrete route choices; cache-heavy long responses expose different runtime bottlenecks from shorter generations; partial collection permits a response to contain tokens sampled under different parameter versions. A method evaluated in one of these regimes may motivate a hypothesis in another, but that transfer is not itself an experimental result.

Appendix A maps every focal study to a primary discussion location. The companion catalog keeps the twenty focal entries separate from foundational and neighboring surveys. It records a paper's editorial role and its evidence type independently. This prevents the size of the bibliography from being mistaken for the number of directly comparable low-precision methods.

6.2. Findings on Stability and Learning Quality

Three recurring findings emerge from the primary studies. Table 3 anchors them in specific comparisons, with source locations recorded in the companion evidence file.

Correction must be evaluated over the learning trajectory. QuRL reports that bounded importance correction can appear adequate early while disagreement grows over a longer run; its ACR and UAQ components address update constraints and quantized-update visibility, respectively [5]. Score Centering isolates a different estimator defect and compares corrections under a shared objective [10]. These observations support monitoring how an intervention changes the update throughout training. They do not establish that every bounded correction fails or that a drift correction reproduces an on-policy trajectory law. AIS and ACRL likewise motivate adaptation or regulation rather than one universally optimal discrepancy target [7, 9].

Numerical coupling is useful at a specified interface. Jet-RL couples precision flow, while QaRL, QUADS, and TRACE couple representations or realized rounding decisions [3, 6, 8, 13]. QUADS's component ablation is especially informative: asymmetric weight QAT recovers much of the learning loss, and rollout-side residual compensation improves the remaining gap. Its comparison of asymmetric and symmetric QAT also shows why nominal symmetry is an inadequate proxy for matched execution. FP8-RL complements these algorithmic studies by documenting the synchronization and calibration needed to keep a changing policy operational [4]. The common result is interface-specific agreement; bitwise identity and transfer to another engine remain separate questions.

Which updates survive can matter as much as average disagreement. The full-pipeline FP8 study connects excessive lower-side clipping of negative-advantage tokens to pathological training dynamics, whereas TRIAGE uses local and direction-dependent information to select interventions [11, 12]. These are distinct routes to controlling the signed learning signal. They explain why an aggregate clipping fraction or mean log-probability gap can miss a relevant mechanism. TRIAGE also limits token-wise directional predictiveness in its MoE routing-conditioned analysis, underscoring the difference between a population pattern and a reliable predictor for each token.

The findings jointly support a conditional conclusion: execution fidelity and learning correction are complementary resources, and the useful balance depends on which failure is active. Evidence is strongest when an internal ablation links a named component to the proposed mechanism and a learning outcome. A reduction in mean discrepancy alone establishes only a numerical observation; absence of collapse establishes stability over the reported horizon; a quality claim additionally requires the stated evaluation and checkpoint rule.

6.3. Efficiency Gains and Implementation Costs

A rollout-only acceleration is not the same as a faster training iteration. A useful accounting decomposition is

$$T_{\text{iter}} = T_{\text{rollout}} + T_{\text{learner}} + T_{\text{sync}} + T_{\text{quant/calib}} + T_{\text{other}}, \tag{11}$$

where terms should be interpreted according to the schedule; overlapping execution requires critical-path accounting rather than naive addition. The equation is a bookkeeping model for serial phases, not a measurement from this survey. It explains why accelerating a dominant kernel can expose another bottleneck.

Different interventions add costs at different interfaces. Repeated scale estimation or numerical conversion can add per-update work, as the FP8-RL integration illustrates [4]. Reference computations can add learner-side overhead. Shared rollout information can require storage, transport, and matching to training examples. Selective precision can add an alternate execution path whose cost

Table 3: Selected published observations and their interpretive limits. These within-paper comparisons illustrate the synthesis; the twenty-work coverage map appears in Appendix A. No row defines a performance ranking across papers.

Study / setting	Published comparison or observation	What the evidence supports
QuRL [5] INT8/FP8 rollout	Longer-run behavior differs from early TIS behavior; ACR and invariant-scaling ablations address distinct failure modes.	Correction magnitude, clipping semantics, and update visibility are separate concerns; a short stable window is insufficient.
QUADS [8] NVFP4 MoE rollout	Weight-only learner QAT recovers much of the loss; adding rollout activation compensation further improves held-out scores.	The two components have complementary evidence within this design; neither universal necessity nor cross-engine transfer follows.
Score Centering [10] BF16 learner; quantization or lag	Shared-objective experiments compare additive centering and importance correction; their composition helps in the reported staleness tests.	Estimator drift and importance-weight control are distinguishable. The implemented behavior expectation is approximated.
Full Pipeline FP8 [11] FP8 learner and rollout	Calibrated Clipping improves the reported learning outcomes across scaling choices. Main GRPO results use best-checkpoint selection over 500 steps.	Restoring signed update contributions can matter; the selected endpoint is not a fixed-budget final-checkpoint comparison.
TRIAGE [12] Native NVFP4	Local tail patterns motivate direction-selective gating and repair. Token-wise predictiveness has a narrower scope than the population characterization.	Local and signed diagnostics can be informative, with limits under MoE coupling; token-independent contraction is not established.
KV4-RL [17] INT4 KV; partial collection	Unprotected INT4 KV misses later learning gains under the retained TIS recipe; bounded INT8 protection recovers near-BF16 endpoint quality.	Conditional correction and cache fidelity address different effects. Timing and quality claims apply to the studied collection workload.

depends on the protected region. These costs should be extracted from the source when available and marked unreported otherwise.

Token throughput also depends on the realized workload. A policy that produces shorter responses can appear faster without improving execution per token; one that produces longer responses can incur more work despite a faster kernel. Memory savings can increase concurrency, which changes the workload seen by attention and compute kernels. Serving research such as PagedAttention provides background for why cache management and scheduling affect observed throughput [23], but serving speed alone does not establish RL learning efficiency.

KV4-RL provides a concrete example of distinct timing scopes: over 400 iterations on Qwen3-8B-Base, it reports 1.80× cumulative rollout speedup and 1.27× cumulative full-iteration speedup over BF16 rollout [17]. The two quantities describe the same study, yet already differ substantially. Its common-BF16 evaluation separates learned-policy comparison from evaluation precision. These figures illustrate timing scope rather than a comparison with another paper's hardware or workload.

The most decision-relevant quantity is often time or compute to a specified quality level. Many reports instead provide endpoint quality, iteration time, or rollout throughput. Those measurements are useful, but they do not permit reconstructing time-to-quality without the corresponding training

Table 4: Information and implementation requirements that distinguish the reviewed approaches. Entries describe method interfaces rather than measured overhead rankings.

Approach	Information or execution dependency	Question for the system comparison
TIS / AIS / CIS	Behavior probabilities and learner/proximal probabilities	Are the stored probabilities those of the actual sampling distribution?
Score Centering	A behavior-weighted score expectation, exact or approximated	What distributional information is returned, and what approximation is used?
Aligned QAT	A learner path corresponding to rollout quantization	Which kernels and numerical interfaces remain different?
Guided rounding	Realized rollout-side quantization information	How are states stored, transferred, and matched to training?
Calibrated Clipping [11]	Higher-precision reference ratios at calibration points	Is the reference-pass cost included in training time?
Selective KV precision	Low-bit cache execution with protected positions	Is protection cost bounded, and under which context workload?

trajectories. Endpoint quality and throughput therefore remain separate observations unless the source reports the training trajectories needed to connect them.

Table 5: Evidence questions for reading efficiency and stability claims. These are literature-extraction fields, not a new benchmark protocol.

Reported claim	Context required	Unsupported shortcut to avoid
Smaller mismatch	Same parameters, prefixes, decoding rules, metric definition	Treating mean gap as a stability guarantee.
Stable training	Horizon, seeds, stopping criteria, quality trajectory	Equating no collapse with useful convergence.
Comparable accuracy	Evaluation precision, tasks, budget, checkpoint selection	Comparing unmatched endpoints as a controlled result.
Faster rollout	Tokens, lengths, concurrency, hardware, baseline	Calling it an equal end-to-end training speedup.
Lower memory	W/A/KV scope, allocated vs resident memory, batch policy	Assuming savings translate directly to throughput.
Method combination	Joint ablation, estimator definitions, additional cost	Assuming different taxonomy classes imply compatibility.

6.4. Limitations of Current Evidence

The literature is heterogeneous enough that a single aggregate score would hide rather than resolve its limitations. Published training horizons, model families, and hardware are not interchangeable. Controlled ablations inside one paper generally support a stronger comparison than absolute numbers across unrelated papers, but an internal comparison still has a finite scope.

There are also limits to what the corpus says about generalization. Mathematical reasoning is a recurring evaluation setting in the collection; broader task claims need corresponding evidence, especially for long interactive trajectories and environments whose state is not simply a token prefix. Experimental success on a specific numerical path should not be extrapolated to a different low-

bit format merely because the bit width matches. Format, scaling, exceptions, and kernel behavior jointly define that path.

Finally, this survey has its own evidence boundary. It organizes twenty focal studies and selected background references, does not reproduce their results, and does not claim exhaustive retrieval. The catalog is intended to make updates and corrections auditable. A missing field is a gap to preserve, not an invitation to infer an unreported implementation detail.

Synthesis. The strongest cross-paper conclusions concern mechanisms and conditions. Numerical agreement, stable optimization, endpoint quality, and runtime savings should remain separate until a source provides evidence linking them.

7. Open Problems

The synthesis suggests three near-term priorities: predicting damage before task scores fall, composing interventions without suppressing useful updates, and documenting behavior policies in stateful or stale execution. The questions below follow those gaps rather than treating every numerical error as equally urgent.

7.1. Predicting Harmful Mismatch

A useful diagnostic would distinguish an innocuous discrepancy from one that will damage learning before the damage is visible in task scores. Current mechanisms suggest several candidates: confidence-conditioned tails, clipping distortions, conditional drift, and directional concentration. The open problem is not simply to propose another statistic, but to establish when it predicts an outcome across settings.

One route is to formulate the prediction target precisely: collapse within a horizon, reduced improvement per update, or failure to reach a quality threshold. These are different labels and may require different predictors. A diagnostic should also be assessed for intervention sensitivity. If a metric falls when a control is applied but learning does not improve, it may track a correlate rather than the damaging mechanism. A decisive result would show predictive value beyond mean mismatch and retain that value after changing the intervention or model. Otherwise, the statistic may identify the original recipe rather than a transferable failure mechanism.

7.2. Composing Numerical and Learning Interventions

Numerical and learning interventions could be coordinated rather than tuned independently. A reduction in execution mismatch may change the appropriate correction strength; a correction that suppresses useful updates may alter the benefit of finer quantization. This suggests studying the marginal value of a component after another component has already changed the operating regime.

The central challenge is to avoid double control. Multiple masks, truncation rules, or trust bands can remove the same examples, while a numerical intervention changes which examples trigger them. A principled composition should expose its target estimator and the surviving learning signal. It should also account for the cost of information exchanged between endpoints. The unresolved quantity is the marginal benefit of a correction after numerical fidelity has improved, measured

alongside retained learning signal and added cost. A combination that only stabilizes a more weakly learning policy would not resolve this question.

7.3. Precision, Staleness, and Long-Horizon Interaction

Asynchronous or partial collection makes the behavior policy token-dependent. Quantization can then interact with version lag: a probability difference may reflect both a changed parameter state and a changed execution path. Equation (1) remains a useful conceptual decomposition, but measuring its factors may require snapshots or replay information that a production system does not retain.

Long responses introduce another challenge. An early sampling change alters the later prefixes at which all subsequent comparisons are made. Per-token correction can be accurate at observed prefixes while the collected trajectory distribution remains different. Adaptive precision may therefore need to account for position, layer, context, and training phase together. The difficult question is where extra fidelity improves learning enough to justify its cost, not merely where it reduces a local tensor error.

A broader extension is multi-turn interaction. Tool observations and environment transitions add sources of trajectory divergence beyond autoregressive token generation. The current taxonomy can still locate numerical and update interventions, but its evidence cannot be assumed to cover these settings. An adequate behavior record may therefore need cache provenance, sampler probability semantics, and external-action history in addition to a checkpoint identifier. Determining the smallest sufficient record is a system and statistical question shared by these settings.

7.4. Generalization and Evidence Standards

The field would benefit from small, explicit evidence contracts that travel with a result. At minimum, a report should identify learner and sampler precision paths, behavior-probability semantics, update-ratio definitions, collection schedule, evaluation precision, and the scope of its timing measurement. These fields are more useful for synthesis than a headline precision label.

Such reporting would also clarify unresolved comparisons. A paper that preserves stability over one horizon could state exactly what remains untested; a system report could distinguish adopted algorithms from its integration contribution; a method paper could specify which computations are simulated and which use native kernels. Maintaining these records in a public reading list would make both positive and negative evidence easier to reuse.

The purpose is not to mandate one universal benchmark. Different studies answer different questions. A shared vocabulary and transparent conditions would allow their results to accumulate without implying that all settings are directly comparable.

8. Conclusion

Low-precision rollout sits at a consequential boundary: the same numerical computation that saves inference cost also supplies the data for learning. Understanding this boundary requires separating execution mismatch from parameter changes, conditional estimation from trajectory selection, and numerical agreement from policy quality.

The component taxonomy developed here organizes existing responses into numerical alignment and fidelity control, estimator correction, update control, and schedule adaptation. It explains why integrated methods occupy more than one category and why format labels alone are insufficient. Across the reviewed work, the useful question is not whether mismatch has disappeared, but which discrepancy has been controlled, what learning signal remains, and under which system conditions the result holds.

A mature account of low-precision RL will need both better numerical interfaces and better evidence about their optimization consequences. The current literature provides concrete mechanisms and promising interventions. Its next step is to make those mechanisms transferable: through explicit assumptions, informative diagnostics, careful composition, and transparent reporting of quality and cost.

A. Focal Corpus and Section Mapping

The following table contains twenty unique focal contributions. A primary location identifies where a work is developed in detail; additional locations identify a distinct component or a contextual use. Supporting roles do not imply absence of direct low-precision experiments. Conference and preprint versions are not counted twice. Foundational and neighboring references are excluded from the count of twenty.

Table 6: Focal works, component classes, and primary discussion locations. Secondary locations are shown in parentheses. All twenty enter the literature coverage synthesis.

Work	Class / role	Location	Scope note
FlashRL / TIS [1]	B1	5.1 (1.1; 2.1)	Quantized rollout; recorded behavior probabilities.
QeRL [2]	Boundary	Box 1	NVFP4 with LoRA and adaptive noise; exploration setting.
Jet-RL [3]	A1; A2	4.1 (6.2)	Unified FP8 flow; granularity is an associated numerical design choice.
FP8-RL [4]	A2; system	4.2 (5.1; 6.2; 6.3)	FP8 stack; adopted TIS/MIS is not a new correction family.
QuRL [5]	B2; A2	5.2 (3.1; 4.2; 5.4; 6.2)	ACR and UAQ are distinct components.
QaRL [6]	A1; B2	4.1 (5.2; 5.4; 6.2)	Aligned QAT and TBPO are distinct components.
AIS [7]	B1	5.1 (6.1; 6.2)	Adaptive estimator mixture; autoregressive and diffusion evidence distinguished.
QUADS [8]	A1; A2	4.1 (4.2; 5.4; 6.2)	Asymmetric QAT plus rollout activation compensation.
ACRL [9]	B2	5.2 (3.3; 6.2)	Advantage-guided, discrepancy-dependent update control.
Score Centering [10]	B1	5.1 (2.2; 3.2; 5.4; 6.2)	Additive score correction; not full on-policy recovery.
Full Pipeline FP8 RL [11]	B2; system	5.2 (3.2; 6.2; 6.3)	Clipping calibration; distinguish from quantizer calibration.
TRIAGE [12]	B2	5.2 (3.3; 6.2)	Direction-aware control under native W4A4 execution.
TRACE [13]	A1	4.1 (3.1; 6.2)	Rollout-guided rounding; local guarantees have limited scope.
MiniRL [14]	Support; B1/B2	2.1 (1.2; 5.1)	Formulation role; also contains direct FP8 rollout experiments.

Continued on next page

Work	Class / role	Location	Scope note
R3 [15]	Support; A1	3.1 (4.1)	MoE routing replay; routing mismatch is not uniquely quantization-induced.
FP16-Mismatch [16]	Support; A2	3.1	BF16/FP16 numerical evidence; not a low-bit acceleration method.
KV4-RL [17]	A2; system	4.2 (5.1; 6.2; 6.3)	INT4 KV with selective protection; colocated partial rollout.
Beyond Precision [18]	Support; B3	5.3 (3.3)	General TIM scheduling evidence; do not infer FP4 validation.
VeXact / Diagnosing TIM [19]	Diagnostic	3.4	Zero-mismatch reference for attribution; not a quantized accelerator.
CIS [20]	Support; B1	5.1 (3.2)	Calibrated importance sampling; BF16 cross-engine evidence.

B. Source Selection and Version Handling

Construction of this snapshot. The starting point was an existing local collection of seventeen papers, expanded by three general TIM studies discussed during scope development. Method and scope statements were checked against available primary papers, official publication pages, and arXiv records. This procedure supports a curated narrative review; it does not supply an exhaustive search history. No screening counts or database coverage claims are inferred retrospectively.

Inclusion and exclusion. A focal study contributes a direct quantized-rollout or low-precision RL intervention, a mechanism needed to interpret that setting, or a carefully identified boundary case. General quantization and RL background is cited selectively. Papers devoted only to low-precision pretraining, unrelated generation modalities, or alternative rollout acceleration do not become focal entries solely because they use the same precision format.

Versions and names. The catalog records the version used where available and preserves first-publication dates separately from publication venues. QeRL, QuRL, and QaRL use the saved formal publication records in the bibliography. Aliases and changes in presentation are not treated as new contributions. Where an arXiv link is used without a pinned version, the catalog distinguishes that link from a locally reviewed version. A subsequent update should revise both the version record and any claim affected by the revision.

Attribution of evidence. Every method-specific statement is attributed to its primary source. Equations (1), (2), (4), (5), and (6) are explanatory identities under their stated conditions; they are not new empirical findings. Figures and summary tables are original organizing devices. The companion repository includes the focal catalog, bibliography, source note, and editorial review record. It contains links to third-party papers rather than redistributed paper files.

C. Interpretation Checklist for Future Updates

1. Identify the actual behavior distribution. Record the parameter version, numerical path, temperature, and sampling truncation; distinguish generated probabilities from learner recomputation.

2. Identify the learner, proximal, and KL-reference roles. Check which ratio a proposed correction or clipping rule actually modifies.
3. Name the intervention component. Use A1/A2/B1/B2/B3 for its direct operation, and record precision target, architecture, and schedule separately.
4. State the remaining approximation. Examples include conditional rather than trajectory-level correction, truncated weights, partial execution alignment, and approximated expectations.
5. Attach each reported outcome to its conditions. Separate numerical fidelity, stability horizon, policy quality, and runtime measurement.
6. Distinguish a published combination from a plausible combination. Keep the latter in the open-problem discussion until direct evidence is available.

These questions support consistent reading and catalog maintenance. They are not a claim that every source already reports every field.

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